

**AI-POWERED TRAIN SAFETY SYSTEM WITH
REAL-TIME CROWD AND OBSTACLE DETECTION**

25-26J-134

Project Proposal Report

Udaya Sri A L S – IT22283412

**B.Sc. (Hons) Degree in Information Technology
Specializing in Data Science**

Department of Information Technology

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

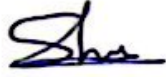
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Table of Contents

DECLARATION	3
ABSTRACT	4
INTRODUCTION	7
BACKGROUND AND LITERATURE SURVEY	8
RESEARCH GAP	9
RESEARCH PROBLEM	10
OBJECTIVES	11
MAIN OBJECTIVE	11
SPECIFIC OBJECTIVES	11
METHODOLOGY	12
OVERALL SYSTEM ARCHITECTURE	12
CROWD DETECTION SYSTEM ARCHITECTURE	13
COMPONENT DEVELOPMENT APPROACH	13
I. <i>Input Processing Development</i>	13
II. <i>Processing Layer Implementation</i>	14
III. <i>Analysis models layer development</i>	15
IV. <i>Integration Layer Implementation</i>	16
V. <i>Output System Development</i>	17
DATA REQUIREMENTS AND STORAGE STRATEGY	17
IMPLEMENTATION TECHNOLOGIES AND TOOLS	18
TESTING AND VALIDATION STRATEGY	19
PROJECT REQUIREMENTS	20
FUNCTIONAL REQUIREMENTS	20
NON-FUNCTIONAL REQUIREMENTS	20
REFERENCES	22

Table of Figures

Figure 1: System Architecture Diagram.....	12
Figure 2: Crowd Detection - High Level System Architecture Diagram.....	13

INTRODUCTION

In Sri Lanka, railway transportation is essential for linking the country's rural and urban inhabitants. It is also a crucial part of national mobility systems. However, the network still faces significant issues including packed cabins, uncomfortable passengers, boarding delays, and safety hazards. During peak hours, these problems are made worse by the unequal distribution of passengers throughout compartments, which leads to inefficiencies and safety risks. By enabling real-time monitoring, analysis, and adaptive management of both passengers and train operations, recent developments in artificial intelligence (AI) and computer vision offer new options to solve these issues [1], [2].

The **Compartment Crowd Density Detection module**, which is the primary focus of this study, applies deep learning-based computer vision techniques to estimate occupancy levels in real time. Models such as YOLO, Faster R-CNN, and CSRNet have demonstrated strong capabilities in detecting and counting individuals under challenging conditions including occlusion, varying lighting, and continuous passenger movement [3], [4]. By providing accurate crowd density insights, the system can direct passengers toward less crowded compartments, reduce congestion, and enhance comfort and safety during travel.

In addition, the Platform Compartment Alignment module dynamically modifies train stopping locations by utilizing the crowd data from both platforms and compartments. This method improves boarding efficiency and reduces harmful crowding at certain doors by aligning congested compartments with less crowded platform zones, in contrast to typical fixed stopping marks [5]. To improve operational safety in a variety of visibility scenarios, the Object Detection system combines data from many sensors to identify obstructions on or close to the tracks [6]. Additionally, to ensure prompt reactions to serious safety hazards, the Intelligent Driver Decision Support system continually assesses the surroundings and sends operators real-time notifications [7]. Few studies have tried to combine disparate solutions—like platform safety management, obstacle detection, or decision support systems—into a cohesive, context-aware framework, despite the fact that prior research has examined these separately. By focusing on crowd density detection and integrating supplementary AI-driven modules, this study pushes the boundaries of the field and provides a flexible and useful solution for the Sri Lankan railway system.

BACKGROUND AND LITERATURE SURVEY

Estimating Passenger Density in Trains through Crowd Counting Modeling

Because uneven crowd distribution poses a safety risk and causes passenger discomfort, crowd density monitoring in train compartments has emerged as a crucial area of research. The accuracy of traditional CCTV-based head-counting techniques is decreased by a few issues, including occlusion, different camera angles, and the presence of luggage. By utilizing computer vision techniques, floor-space occupancy prediction and recent methods go beyond basic passenger counting and offer real-time assistance via intelligent passenger information systems and digital signs. [8]

Efficient Solution to Avoid Overcrowding in Local Train Compartments

Accurate compartment-level density estimation has emerged as a key area of research interest since overcrowding and unequal passenger distribution compromise local trains' safety and comfort. The influence of luggage on useable space and the real-time spatial distribution of standing crowds are not captured by earlier sensor-based methods, such as ToF entry-exit counts, which can only estimate the total number of people. Similarly, occlusion, changing lighting, and limited camera viewpoints limit the reliability of classic CCTV head-counting techniques. Recent computer vision-driven methods, on the other hand, use temporal crowd modeling, free-space occupancy prediction, to enable dynamic passenger assistance through mobile applications and intelligent signs. [10]

Towards Better Railway Service: Passengers Counting in Railway Compartment

Conventional compartment-level passenger counting techniques, such as the motion-supervised head identification used in the railway project, ignore luggage volume, object placement, and station-wide crowd dynamics in favor of concentrating only on passenger enumeration within isolated compartments. By combining station-wise analysis with compartment-specific density evaluation and employing CCTV networks to concurrently identify passenger, luggage, and available space volumes, modern crowd analysis systems get beyond these restrictions. By offering operators intelligence for proactive passenger routing and safety optimization based on actual space usage rather than headcount data, this advancement takes railway operations from simple counting to responsive crowd management. [9]

RESEARCH GAP

Intelligent train compartment population density detection still has a number of significant shortcomings, despite advancements in smart transportation and crowd monitoring.

I. Lack of Predictive Analytics for Passenger Flow

Current solutions often provide only static crowd status. They do not incorporate predictive models to forecast platform occupancy at upcoming station using historical and real-time data.

II. Limited Multi-Model Data Fusion

Without including historical boarding trends, the majority of studies mostly depend on single-source data (such as CCTV feeds), which lowers reliability in congested or obscured situations.

III. Limited Information Flow and Passenger Engagement

Few systems use mobile applications to give passengers with real-time crowding information, even if some offer operator dashboards. This disparity lowers overall trip experience by preventing travelers from selecting the best compartment.

By creating a multi-source, AI-powered decision support system that integrates computer vision, predictive modeling, and real-time interface interaction with passenger interfaces, our research fills these gaps. In order to improve passenger comfort, safety the system seeks to deliver precise, flexible, and useful crowd density information.

RESEARCH PROBLEM

To enhance travel comfort, safety, how can we create a real-time intelligent train crowd monitoring and optimization system that precisely measures the passenger density by compartment, calculates usable free space?

Crowd Detection Challenge

Using camera feeds, accurately identify passengers and baggage in real time while avoiding problems with occlusion, bad camera angles, and inconsistent illumination within compartments.

Free-space Estimation Challenge

calculating a compartment's useable space by taking into consideration factors such as baggage placement, passenger density, and unequal crowd distribution instead of depending just on the number of passengers.

Passenger Communication Challenge

By ensuring that passengers receive real-time crowding notifications through mobile applications, overall comfort and safety are improved and informed compartment selection is made possible.

OBJECTIVES

Main Objective

To improve passenger safety, maximize boarding efficiency, and deliver timely operational guidance within the Sri Lankan railway network by creating an AI-powered Train Safety and Passenger Guidance System that combines real-time object detection, crowd density estimation and prediction, driver decision support, and platform-compartment alignment.

Specific Objectives

SO1: Real-Time Compartment Crowd Detection

Develop and train machine learning models capable of predicting optimal hardware allocation based on code analysis features, achieving minimum 85% accuracy in hardware recommendation for standard benchmark problems.

SO2: Free Space and Occupancy Estimation

Provide exact metrics for occupancy levels and comfort prediction by designing algorithms that compute useable compartment capacity (free space) while taking into consideration passenger density, baggage, and uneven crowd distribution.

METHODOLOGY

Overall System Architecture

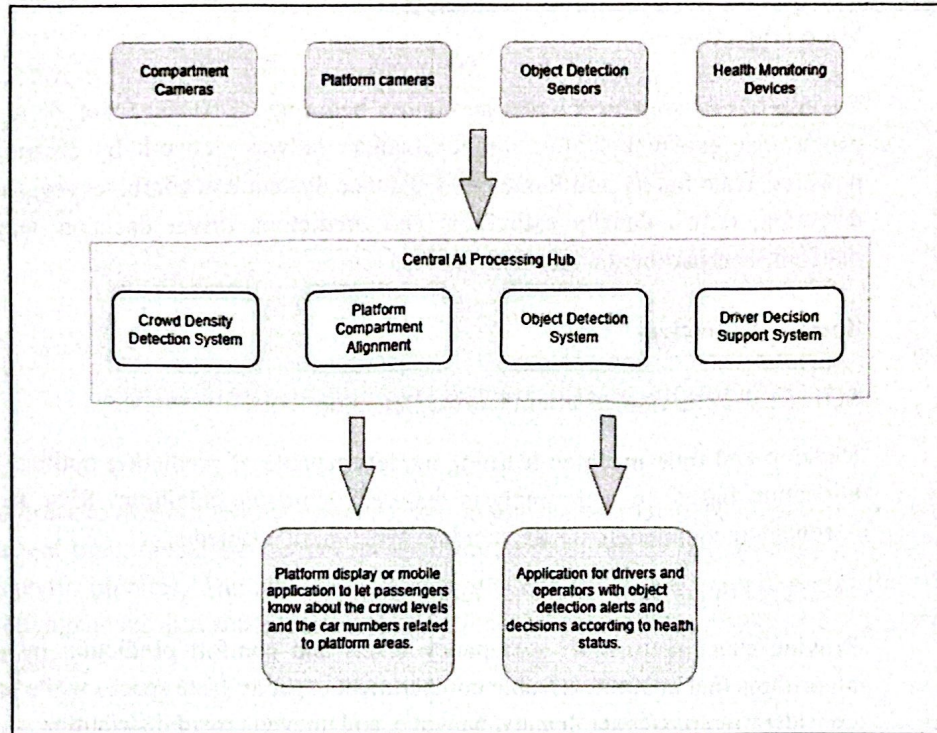


Figure 1: System Architecture Diagram

The Railway Crowd Detection and Management system's key intelligence component, the key AI Processing Hub, receives real-time data from Platform Cameras, Compartment Cameras, Object Detection Sensors, and Health Monitoring Devices, as seen in Figure 1.

Four main systems are integrated by the Hub: Platform Compartment Alignment optimizes distribution, Object Detection detects passengers and luggage, Driver Decision Support offers operational advice, and Crowd Density Detection assesses passenger occupancy.

Users are served by two output streams: operator-facing apps provide drivers detection alerts and make decisions based on real-time health monitoring for increased safety, while passenger-facing platform displays and mobile applications provide information on crowd levels and the best places to board

Crowd Detection System Architecture

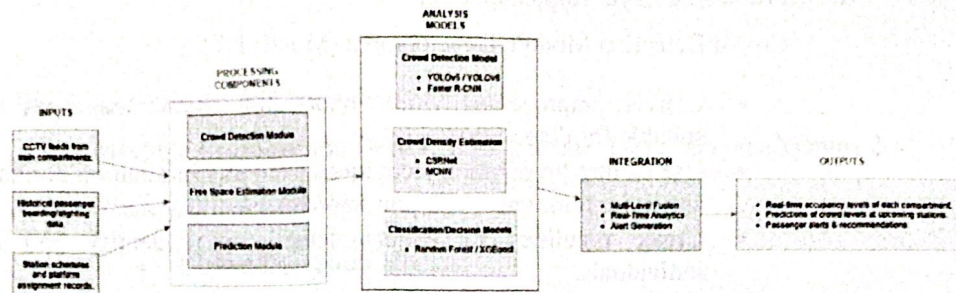


Figure 2: Crowd Detection - High Level System Architecture Diagram

In order to monitor the passenger density in train compartments and between stations in real time, the suggested system architecture functions as a multi-stage intelligent analytic pipeline. The system utilizes sophisticated ML/DL models and combines heterogeneous data sources to produce actionable insights for passengers as shown in Figure 2.

Component Development Approach

I. Input Processing Development

Input Component Integration (Month 1)

- **CCTV Feed Interface:** Create APIs to capture live video feeds with frame rate optimization and quality control from train compartments, platforms, and station entrances.
- **Monitor of Historical Data:** Put in place real-time tracking of passenger boarding and alighting habits, seasonal trends, changes in peak hours, and the effects of special events.

II. Processing Layer Implementation

Crowd Detection Module Development (Month 1-2)

- Convert unprocessed video inputs into frame sequences that are suitable for detection.
- Use picture preparation techniques such as illumination normalization and noise removal.
- Create pipelines for object detection to identify and monitor individuals.
- Develop methods for extracting features from crowd density indicators.

Free-Space Estimation Module Development (Month 2)

- Calculate free volume in compartments using ML/DL model integration.
 1. Passenger Count Estimation:
 2. Passenger Occupied Volume Calculation
 3. Compartment Capacity Integration
 4. Free Volume Computation
 5. Density-Based Refinement

- Integration of Space Allocation and Luggage Detection

Development of Prediction Modules (Month 2)

- Create frameworks for time series preprocessing.
- Use feature engineering to determine the patterns of passenger flow.
- Develop techniques for extracting temporal patterns.
- Create input validation to ensure compatibility with past data.

Implementation of the Passenger Interface (Month 2)

- Create the ability to broadcast data in real time. (via mobile app for passengers).
- Put access control and user authentication measures into place.
- Establish systems for delivering notifications.

III. Analysis models layer development

Machine Learning and Deep Learning Model Development (Months 2-4)

Phase I: Model Architecture

- Crowd Detection Models:
 - Implementation of YOLOv5/YOLOv8 for real-time person detection.
 - Rapid R-CNN integration for crowd recognition with high accuracy.
 - Multi-scale detection for different density of crowds.
- Crowd Density Estimation :
 - Implementation of CSRNet for precise density mapping.
 - Multi-column crowd counting with MCNN development.
 - Mechanisms of confidence grading for density forecasts

Phase II: Model Training and Optimization (Month 3-4)

Month 3: Core model training

Computer Vision Models Training:

- Create training datasets using generated data and annotated crowd scenarios.
- Train YOLOv5/YOLOv8 for crowd detection with hyperparameter tuning.
- Train MCNN and CSRNet to estimate crowd density.
- Conduct cross-validation to verify the resilience of the model.
- Create algorithms for accurately predicting crowd density.

Month 4: Advanced Model Training Development

The Creation of Hybrid Models:

- YOLO + LSTM: Develop and hone hybrid models that include temporal prediction and real-time detection.

- Testing the integration of time series and computer vision components.
- Validation of cross-models and performance comparison.

Training Classification/Decision Models:

- Random Forest: Put ensemble classifiers into practice and train them for crowd-level classification.
- XGBoost: Create and refine gradient boosting models to maximize efficiency.
- Trees of Decision: For operator guidance, develop and train interpretable decision routes.
- Use and train support vector machines for tasks involving binary categorization (safe/unsafe crowd levels).

Validation and Optimization of the Model:

- thorough assessment of model performance for every kind of model.
- Fine-tuning hyperparameters for peak performance.
- Model selection according to resource needs, speed, and accuracy.

Phase III: Model Integration and System Optimization (5-6 Month)

- Create ensemble combination strategies
- System-Level Integration:

IV. Integration Layer Implementation

Multi-Level Fusion Development (Months 5-6)

- Use a weighted decision-making process based on model accuracy ratings for confidence scoring.
- Conflict Resolution: Create methods that settle disputes across several models of analysis.
- The last suggestion is: Construct thorough crowd management synthesis systems.

Real-Time Analytics Engine

- **Stream Processing:** Create pipelines for high-throughput data processing.
- **Pattern Recognition:** Use anomaly detection to identify odd population dynamics.
- **Develop forward-looking crowd projection systems** using predictive analytics.
- **Monitoring Performance:** Construct systems for tracking system health and correctness.

V. Output System Development

Output Components (Month 6)

Real-time Occupancy Reporting

- Create live occupancy figures for the compartments along with confidence intervals.
- Forecast the anticipated audience density for future stations.

Crowd Level Predictions

- Create short-term crowd predicting algorithms for the next 15 to 30 minutes.
- Put in place medium-term demand forecast tools for the next two to four hours.
- Make long-term forecasts of the daily or weekly passenger flow.

Data Requirements and Storage Strategy

Data Collection Strategy

I. Training Data Sources:

Video Data Collections

- CCTV footage captured in high definition from platforms, station entrances, and train compartments.

Historical Passenger Data

- Two to three years' worth of Sri Lanka Railways database boarding/alighting records.
- Patterns of seasonal passenger movement, such as during the monsoon, holidays, and festivals.
- Information about special events (public holidays, religious festivals, and cricket matches).
- Records of past cancellations and delays together with the accompanying effects on crowds.

Real-time Operational Data

- Real-time train timetables, platform assignments, and interruptions in service.
- Correlation between passenger flow patterns and weather data.
- Data on ticket sales and trends in advance reservations.
- Infrastructure design and standards for station capacity.

Implementation Technologies and Tools

Core Technologies

- Python 3.9+ with specific computer vision and machine learning libraries are examples of programming languages.
- Computer vision frameworks for image processing and video analysis include PIL/Pillow and OpenCV 4.5+.
- Machine Learning: TensorFlow 2.8+/PyTorch 1.12+ for ensemble techniques and deep learning, scikit-learn, XGBoost
- YOLOv5/YOLOv8 (Ultralytics) and Detectron2 (Facebook Research) for Object Detection for Quicker R-CNN Implementation
- Crowd Analysis: TensorFlow/PyTorch-based custom implementations of CSRNet and MCNN
- Deep Learning: PyTorch and Keras Lightning for the creation of LSTM/Bi-LSTM/GRU models.
- Real-time processing: Redis for real-time data caching and session management, Apache Kafka for data streaming
- Mobile Development: Flutter as a substitute framework, React Native for cross-platform mobile apps

- **Data management:** MongoDB for video information and annotations, PostgreSQL for historical passenger data

Testing and Validation Strategy

Testing Framework

- **Unit Testing:** Individual component validation with 90%+ code coverage across all layers.

PROJECT REQUIREMENTS

Functional Requirements

Video Input Processing

- Record live CCTV footage from train compartments while minimizing noise and optimizing frame rate.
- Preprocess frames (denoising, normalization of lighting).

Passenger and Luggage Detection

- Real-time passenger detection and counting is achieved through the use of deep learning models (YOLO, Faster R-CNN).
- To distinguish between usable and occupied space, identify the bags.

Crowd Density Detection

- Estimate the passenger density per compartment in real time by using crowd counting models (CSRNet, MCNN).
- Determine the amount of usable free space while taking luggage placement and passenger distribution into account.

Real-Time Output and Notifications

- Show available space and occupancy levels on passenger-facing applications (platform screens, mobile devices).

Non-Functional Requirements

Performance and Efficiency

- processing video feeds in real time with a latency of less than two seconds.
- Manage several video feeds at once without experiencing system crashes.

Accuracy and Reliability

- Passenger detection accuracy $\geq 90\%$ under varied conditions (occlusion, lighting changes).
- Crowd density estimation accuracy $\geq 85\%$.
- Fault-tolerant prediction models with fallback mechanisms.

Scalability

- Integration of cloud and edge for extensive implementation.

Usability

- Simple, color-coded indications (e.g., Green = Free, Yellow = Moderate, Red = Crowded) should be used to represent occupancy statistics on mobile and display interfaces.

Security and Privacy

- Verify CCTV data anonymization (no storing of facial recognition).
- Secure data transmission between apps, processing hub, and cameras.

Maintainability & Extensibility

- New datasets should be compatible with models without necessitating a complete system redesign.
- In order to integrate new AI models, such as YOLOv8 in place of YOLOv5, the design is modular.

REFERENCES

- [1] Li, H., Chen, J., & Chen, S. (2021). "Deep learning applications in intelligent transportation systems." *IEEE Transactions on Intelligent Transportation Systems*.
- [2] Ranjan, R., et al. (2020). "AI-driven solutions for railway safety management: A review." *Transportation Research Part C*.
- [3] Zhang, Y., Zhou, D., Chen, S., Gao, S., & Ma, Y. (2016). "Single-image crowd counting via multi-column CNN." *CVPR*.
- [4] Redmon, J., & Farhadi, A. (2018). "YOLOv3: An incremental improvement." *arXiv preprint arXiv:1804.02767*.
- [5] Jiang, T., et al. (2019). "Passenger flow management in metro systems using real-time video analytics." *Journal of Rail Transport Planning & Management*.
- [6] Li, Z., et al. (2021). "Railway obstacle detection using multi-sensor fusion under low visibility." *IEEE Access*.
- [7] Kumar, A., & Singh, R. (2020). "AI-based decision support systems for train operators." *Procedia Computer Science*.
- [8] B. Tjandra, O. J. Jodrian, N. S. C. Handoko, and A. F. Wicaksono, "Estimating Passenger Density in Trains through Crowd Counting Modeling," *Jurnal Ilmu Komputer dan Informasi (Journal of Computer Science and Information)*, vol. 18, no. 1, pp. 67-76, 2025, doi: 10.21609/jiki.v18i1.1314.
- [9] Y. Liang, X. Qian, and L. Zhu, "Towards Better Railway Service: Passengers Counting in Railway Compartment," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 31, no. 2, pp. 439-[end page], Feb. 2021.
- [10] M. Jambhale, S. Joshi, S. Amrutkar, and N. Baisane, "Efficient Solution to Avoid Overcrowding in Local Train Compartments," in *Proc. 2020 Fifth Int. Conf. Research in Computational Intelligence and Communication Networks (ICRCICN)*, 2020.